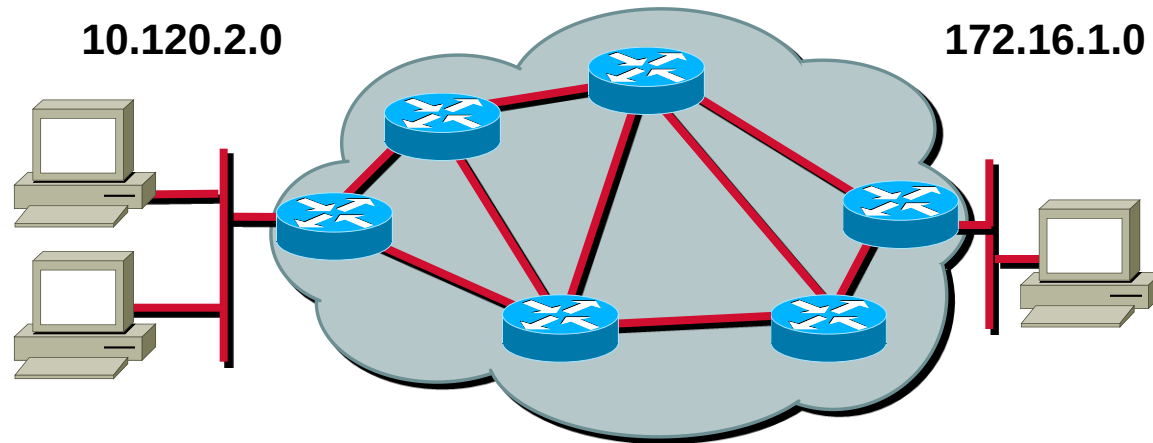


Objectives

Upon completion of this chapter, you will be able to complete the following tasks:

- **Distinguish the use and operation of static and dynamic routes**
- **Configure and verify a static route**
- **Identify how distance vector IP routing protocols such as RIP operate on Cisco routers**
- **Enable Routing Information Protocol (RIP)**
- **Verify IP routing with show and debug commands**

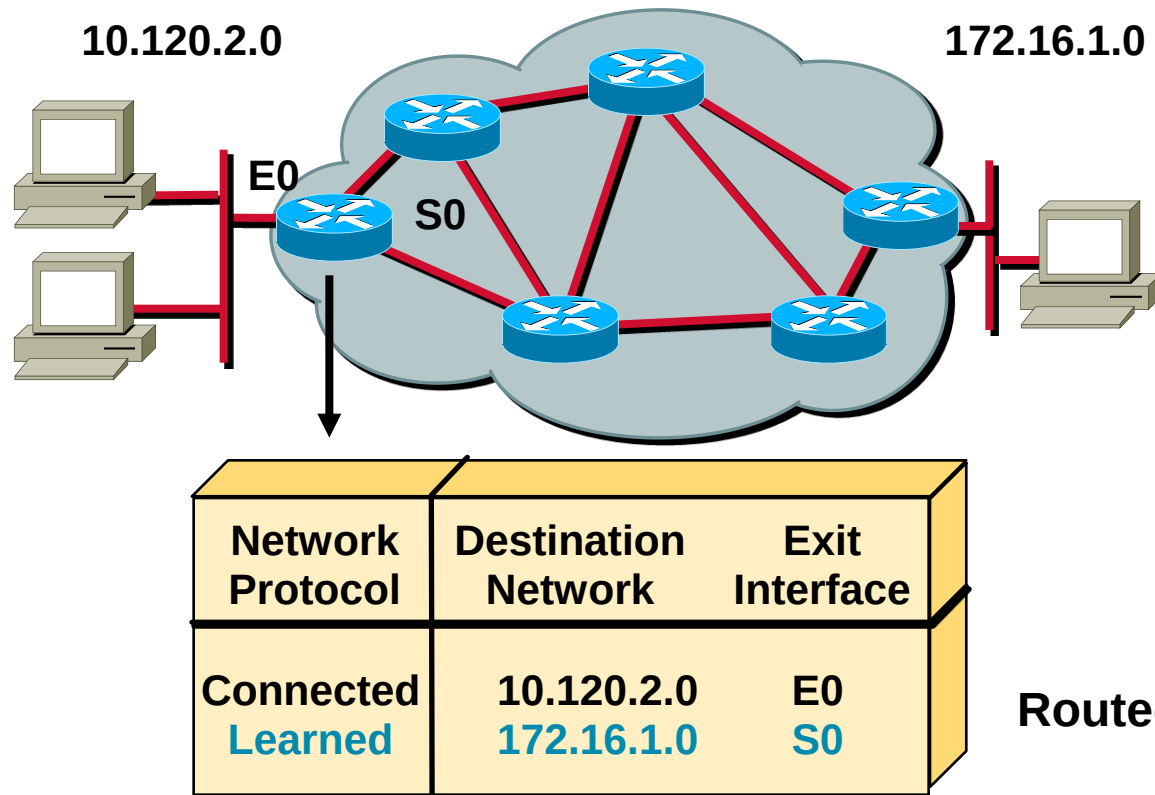
What is Routing?



To route, a router needs to know:

- Destination addresses
- Sources it can learn from
- Possible routes
- Best route
- Maintain and verify routing information

What is Routing? (cont.)



Routers must learn destinations that are not directly connected

Identifying Static and Dynamic Routes

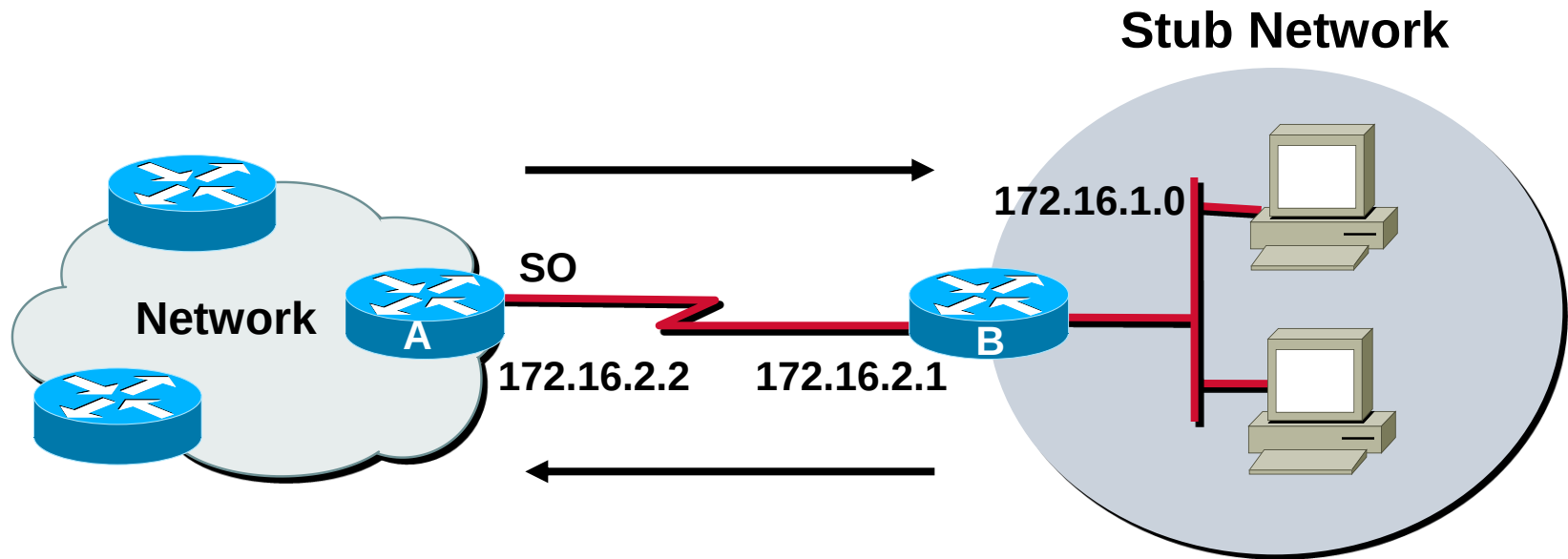
Static Route

Uses a route that a network administrator enters into the router manually

Dynamic Route

Uses a route that a network routing protocol adjusts automatically for topology or traffic changes

Static Routes



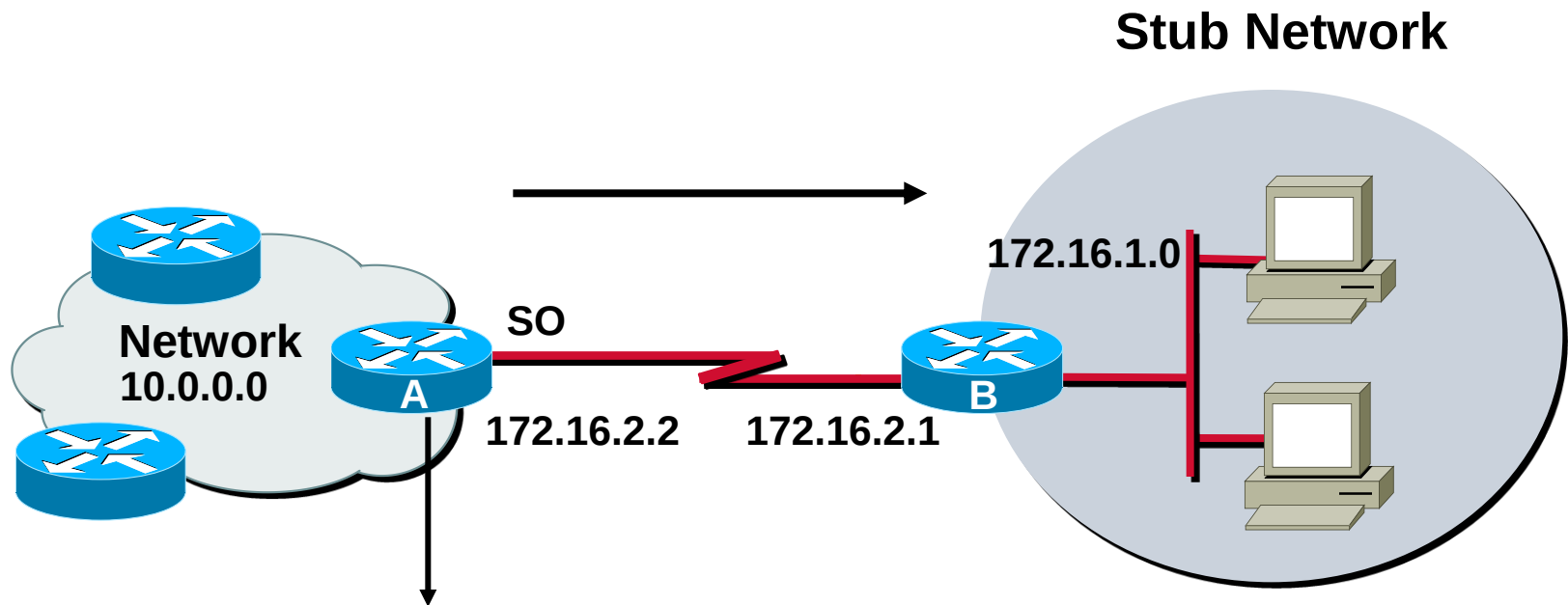
Configure unidirectional static routes to and from a stub network to allow communications to occur.

Static Route Configuration

```
Router(config)#ip route network [mask]  
{address | interface} [distance] [permanent]
```

Defines a path to an IP destination network or subnet

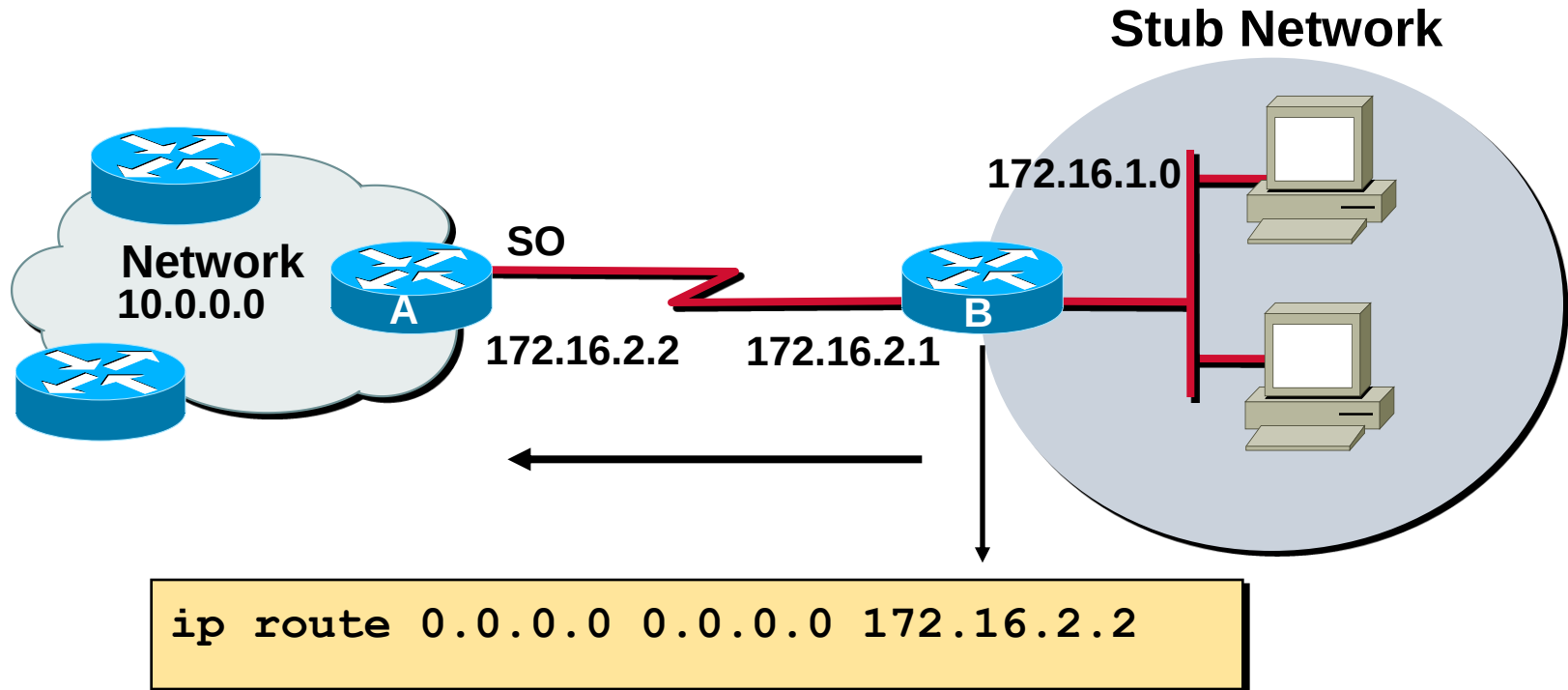
Static Route Example



```
ip route 172.16.1.0 255.255.255.0 172.16.2.1
```

- This is a unidirectional route. You must have a route configured in the opposite direction.

Default Routes

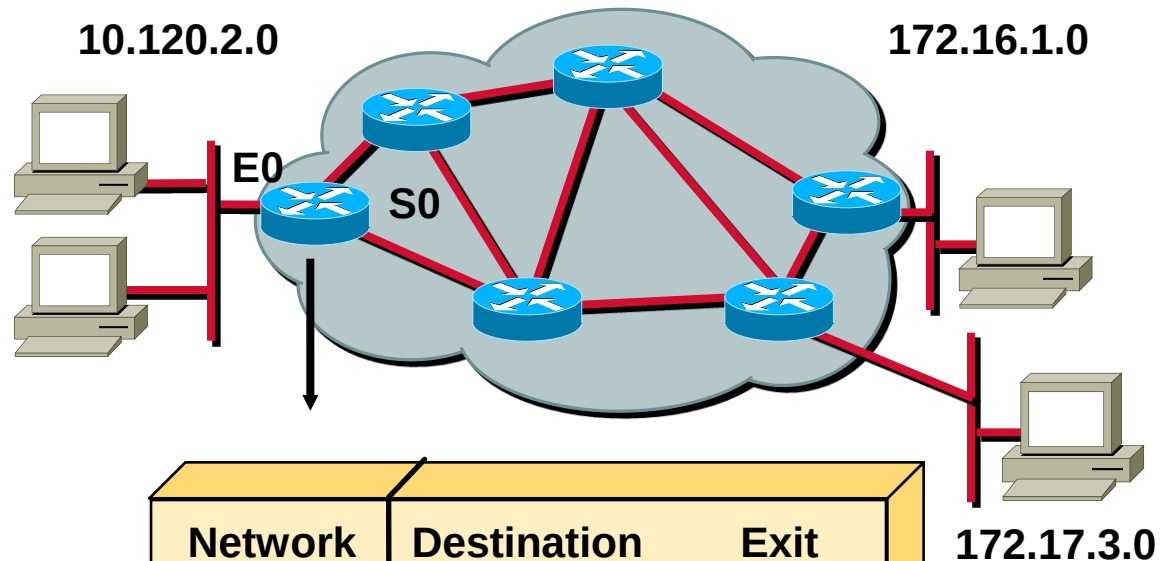


- This route allows the stub network to reach all known networks beyond router A.

What is a Routing Protocol?

Routing protocols are used between routers to determine paths and maintain routing tables.

Once the path is determined a router can route a **routed** protocol.



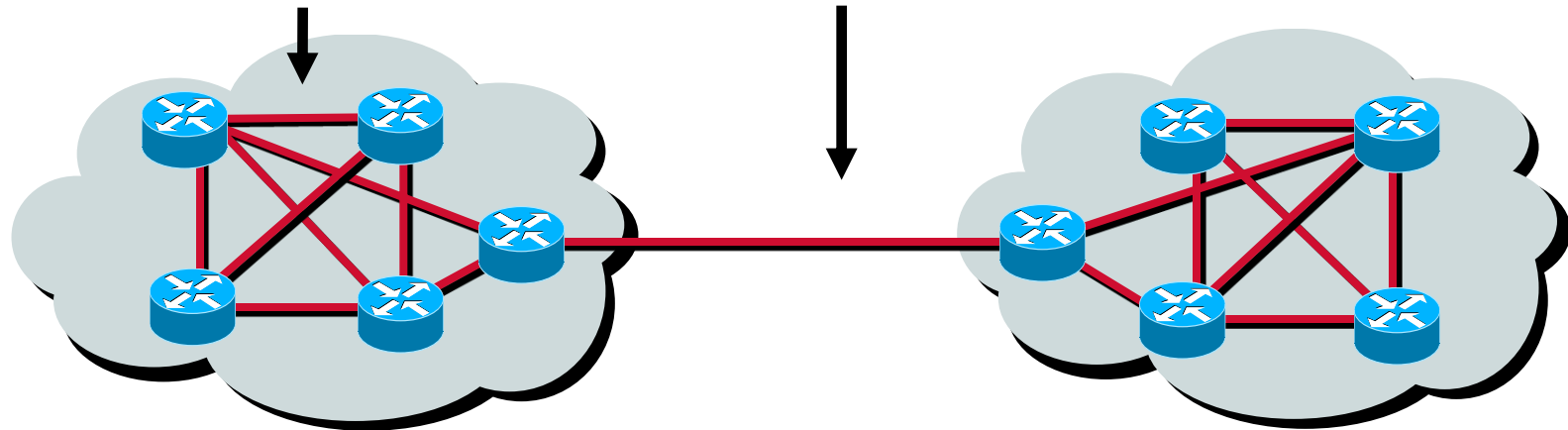
Network Protocol	Destination Network	Exit Interface
Connected RIP IGRP	10.120.2.0	E0
	172.16.2.0	S0
	172.17.3.0	S1

Routed Protocol: IP
Routing protocol: RIP, IGRP

Autonomous Systems: Interior or Exterior Routing Protocols

IGPs: RIP, IGRP

EGPs: BGP

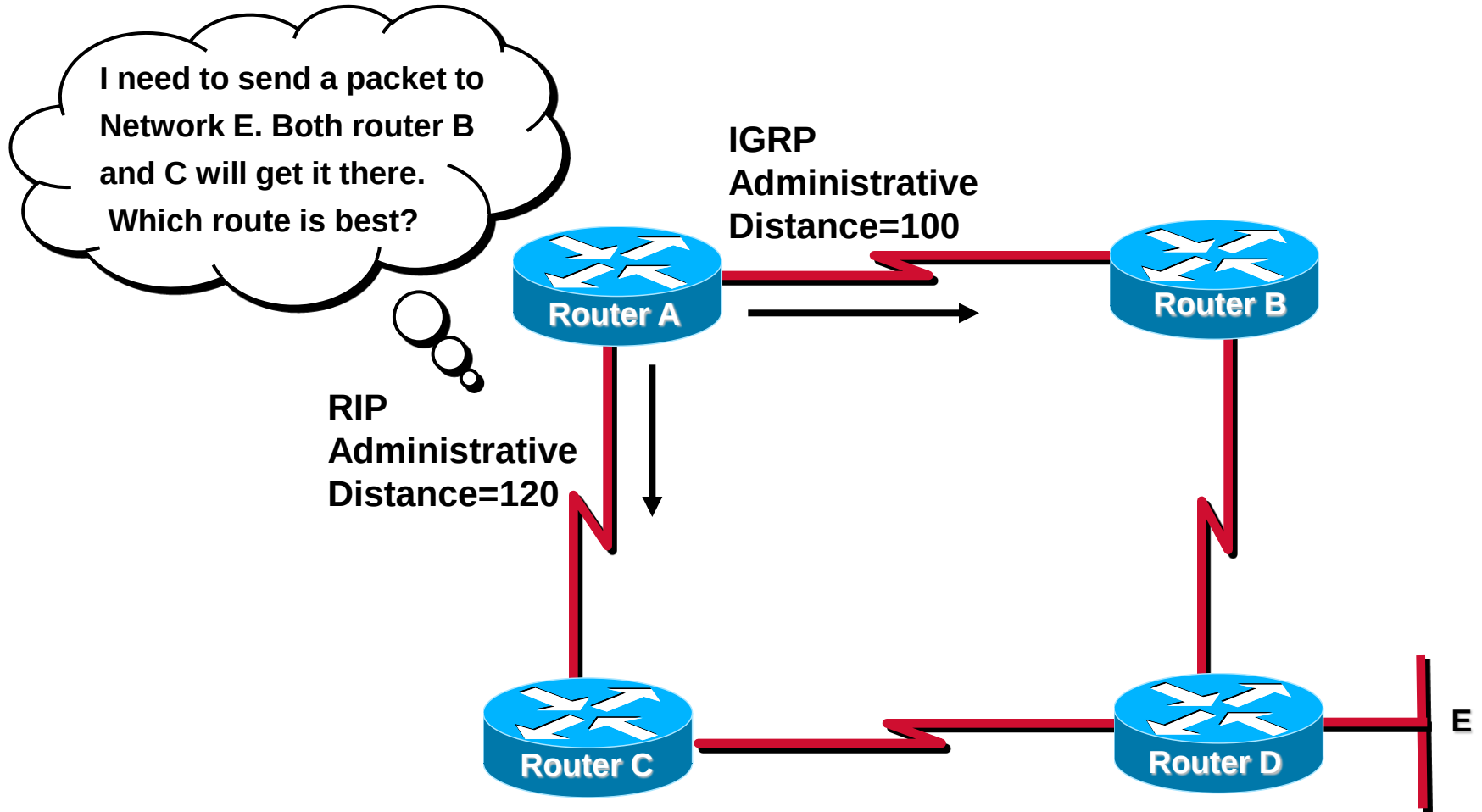


Autonomous System 100

Autonomous System 200

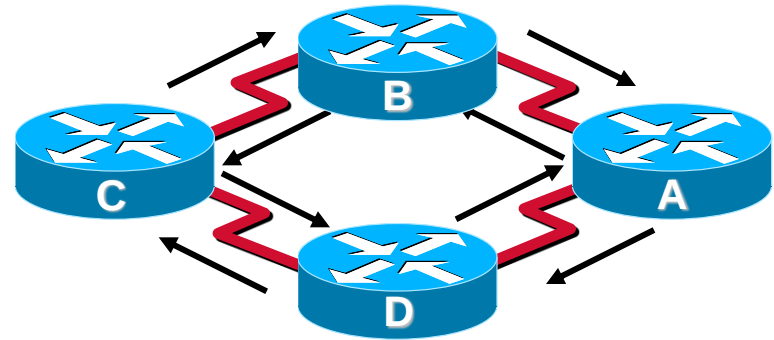
- An autonomous system is a collection of networks under a common administrative domain
- IGPs operate within an autonomous system
- EGPs connect different autonomous systems

Administrative Distance: Ranking Routes

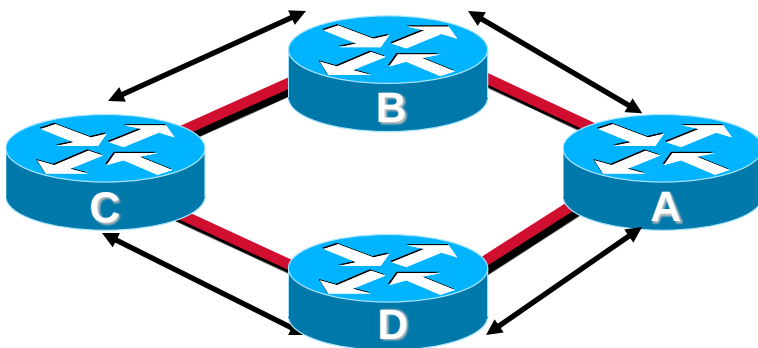


Classes of Routing Protocols

Distance Vector

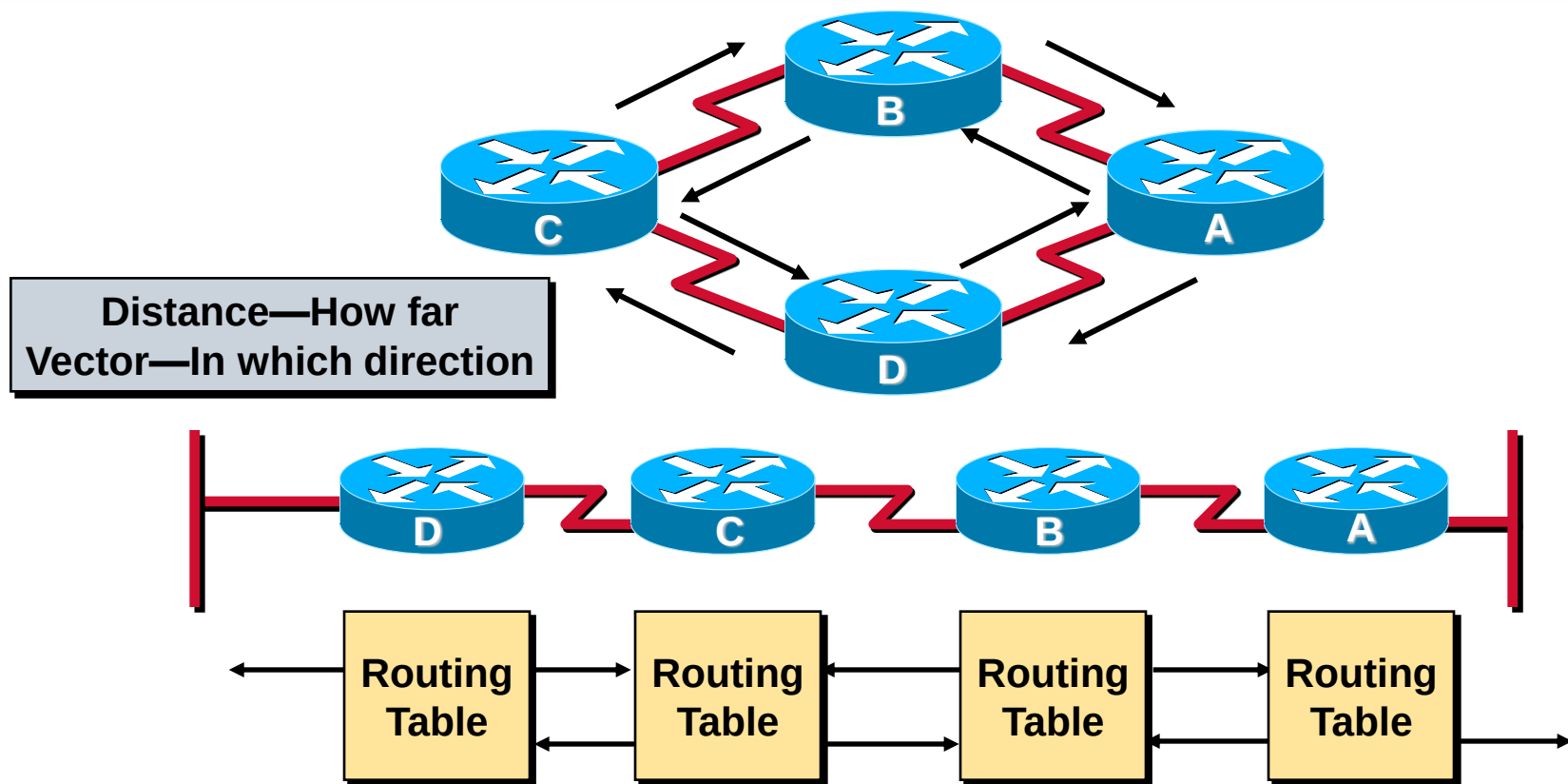


Hybrid Routing



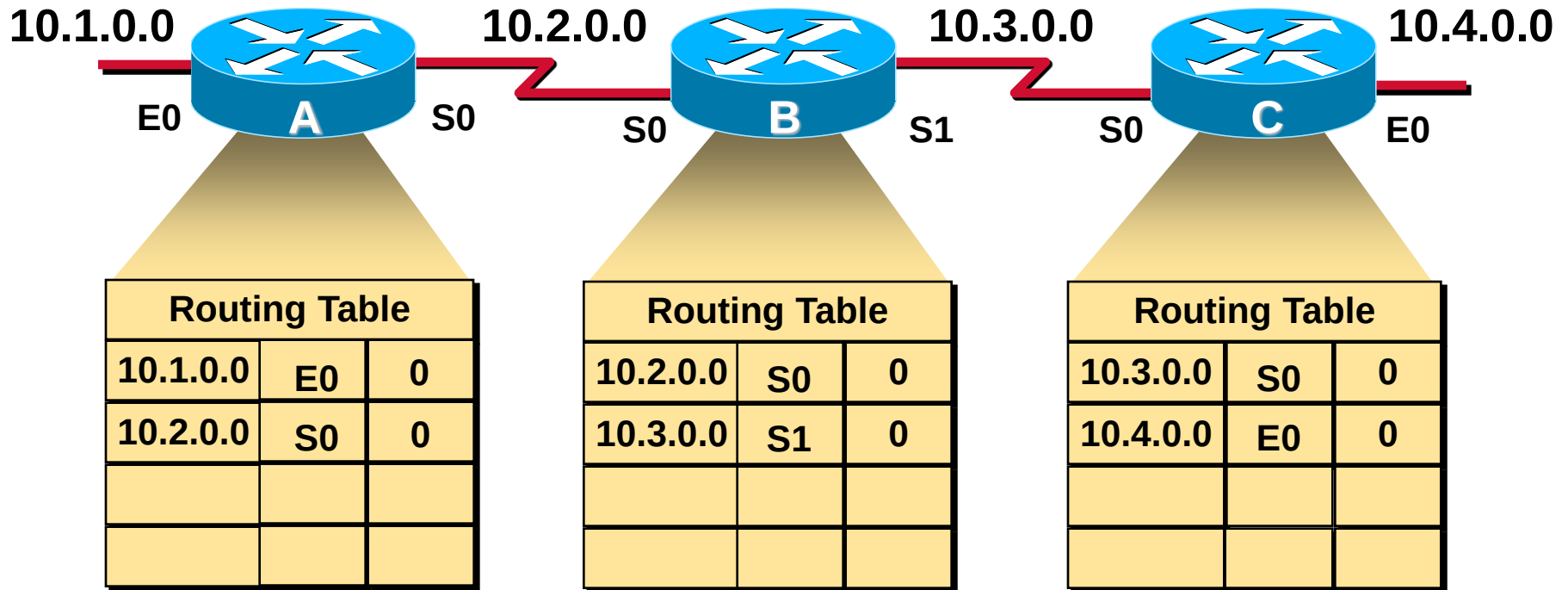
Link State

Distance Vector Routing Protocols



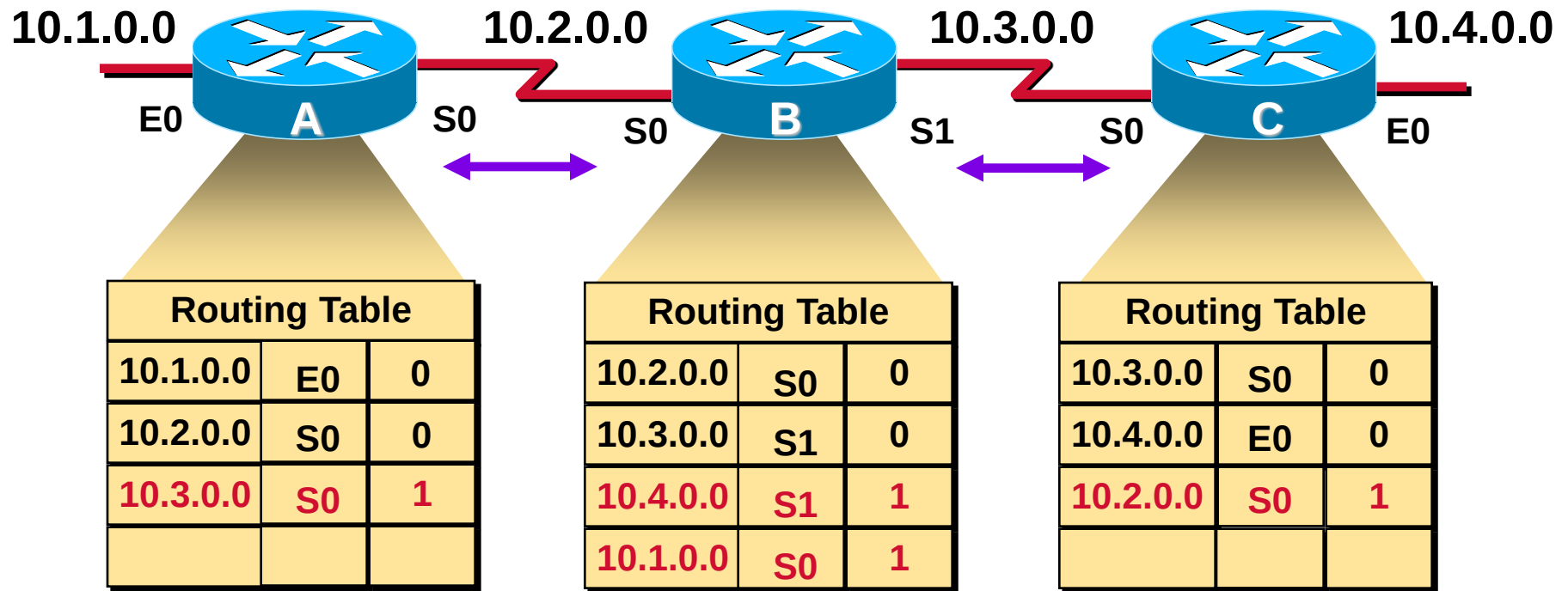
Pass periodic copies of routing table to neighbor routers and accumulate distance vectors

Distance Vector—Sources of Information and Discovering Routes



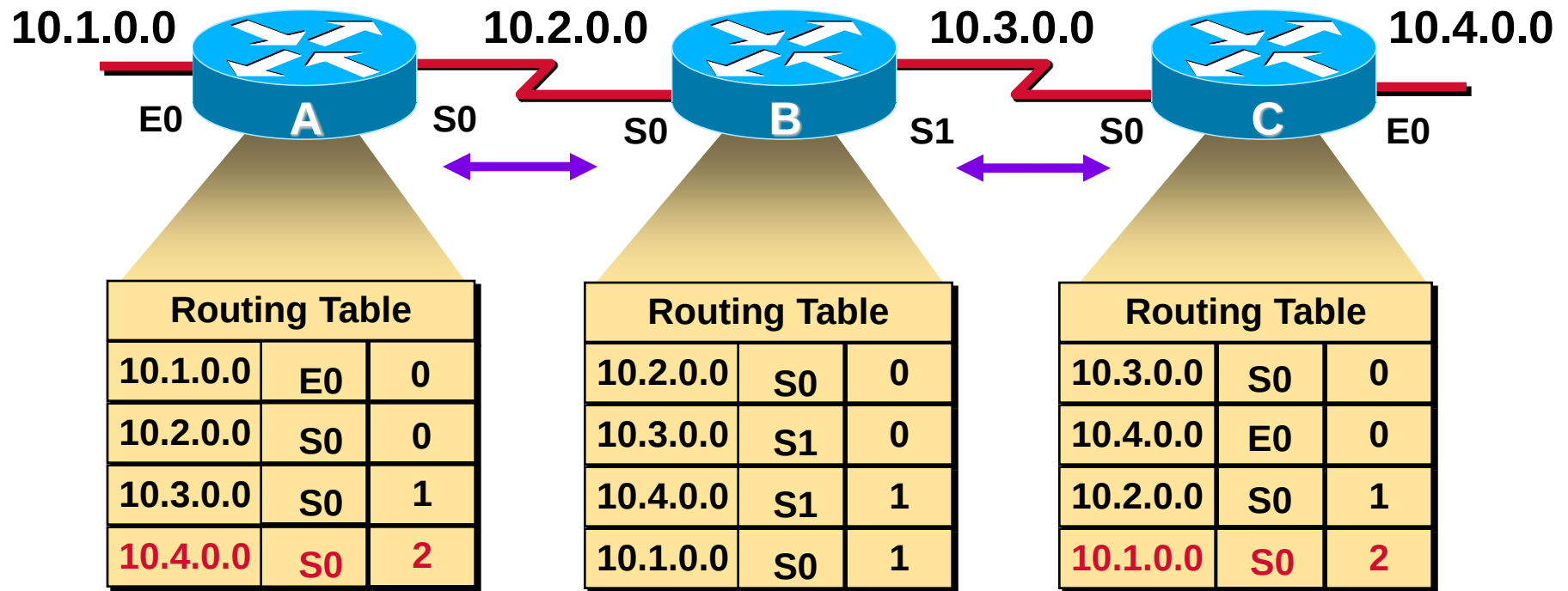
Routers discover the best path to destinations from each neighbor

Distance Vector—Sources of Information and Discovering Routes



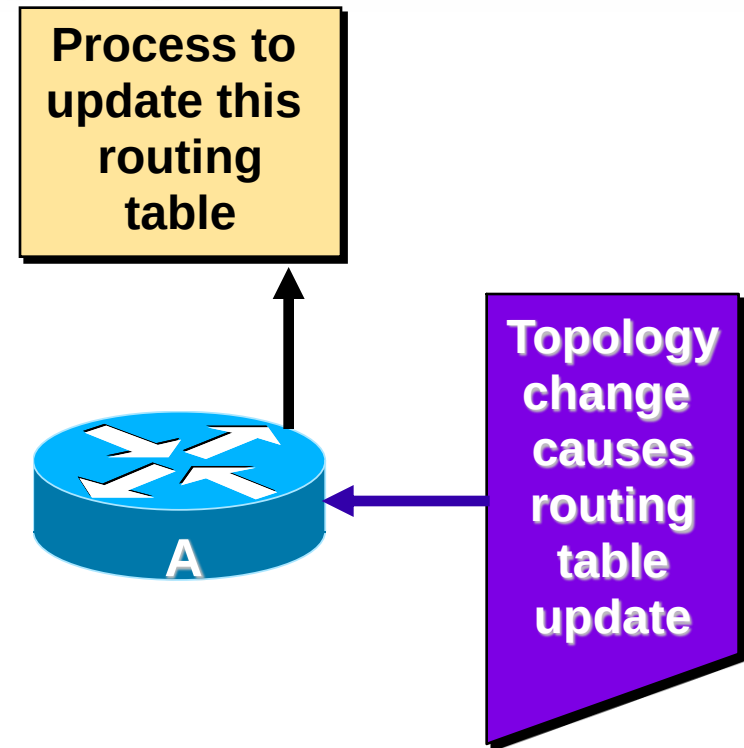
Routers discover the best path to destinations from each neighbor

Distance Vector—Sources of Information and Discovering Routes



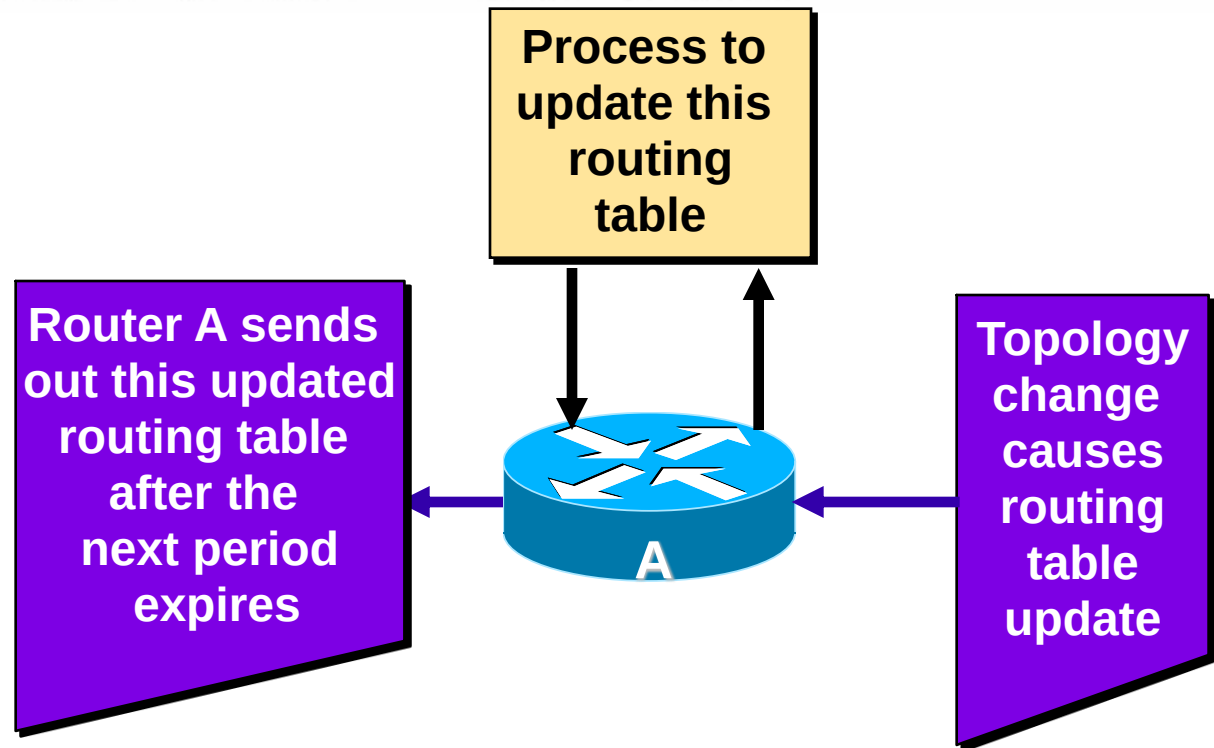
Routers discover the best path to destinations from each neighbor

Distance Vector—Maintaining Routing Information



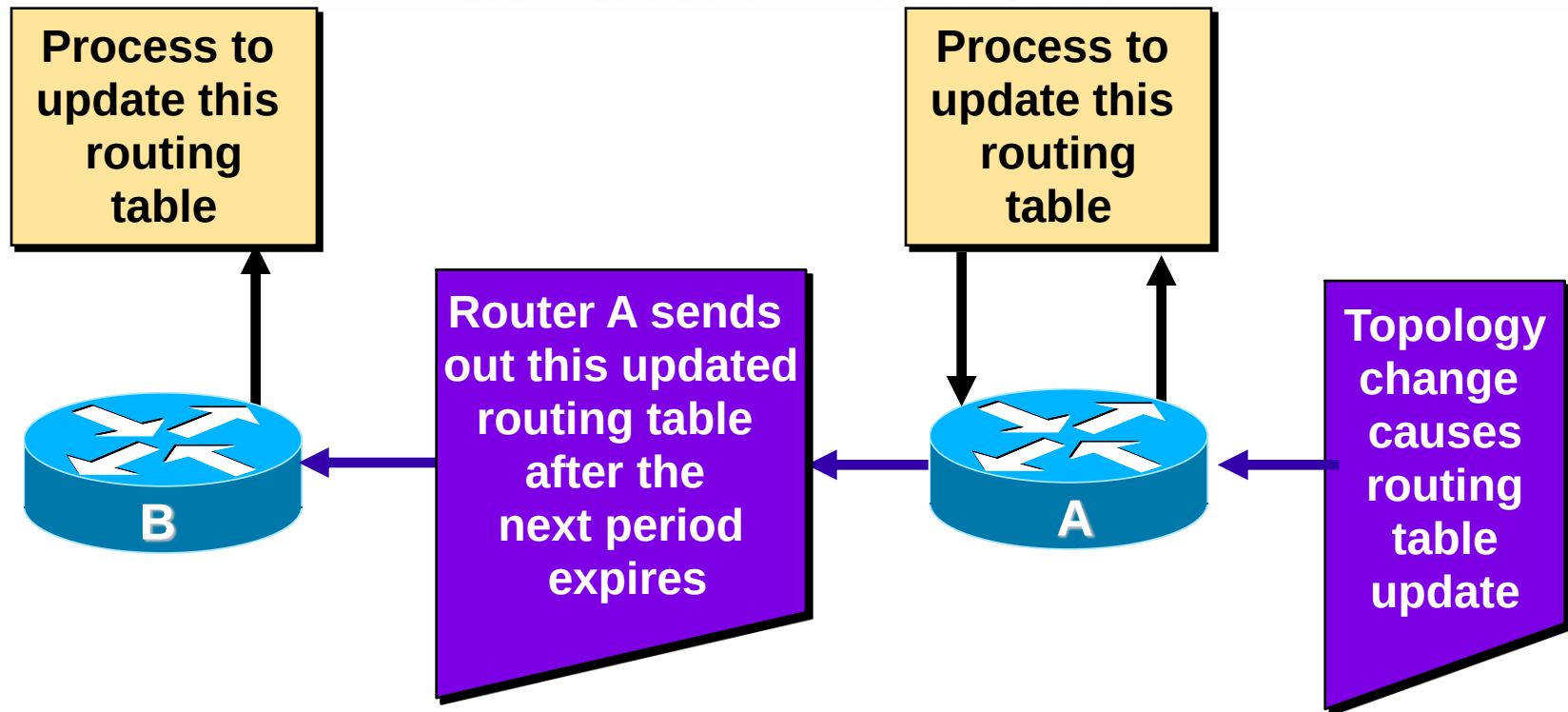
**Updates proceed step-by-step
from router to router**

Distance Vector—Maintaining Routing Information



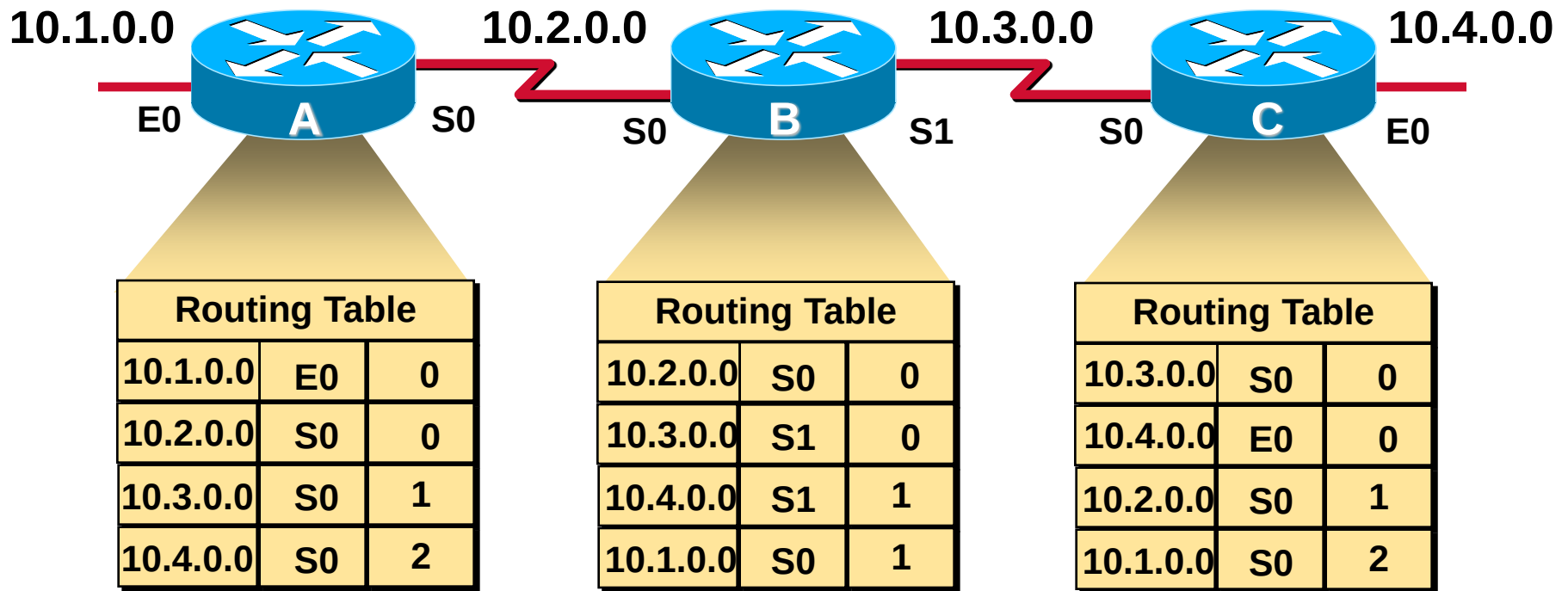
**Updates proceed step-by-step
from router to router**

Distance Vector—Maintaining Routing Information



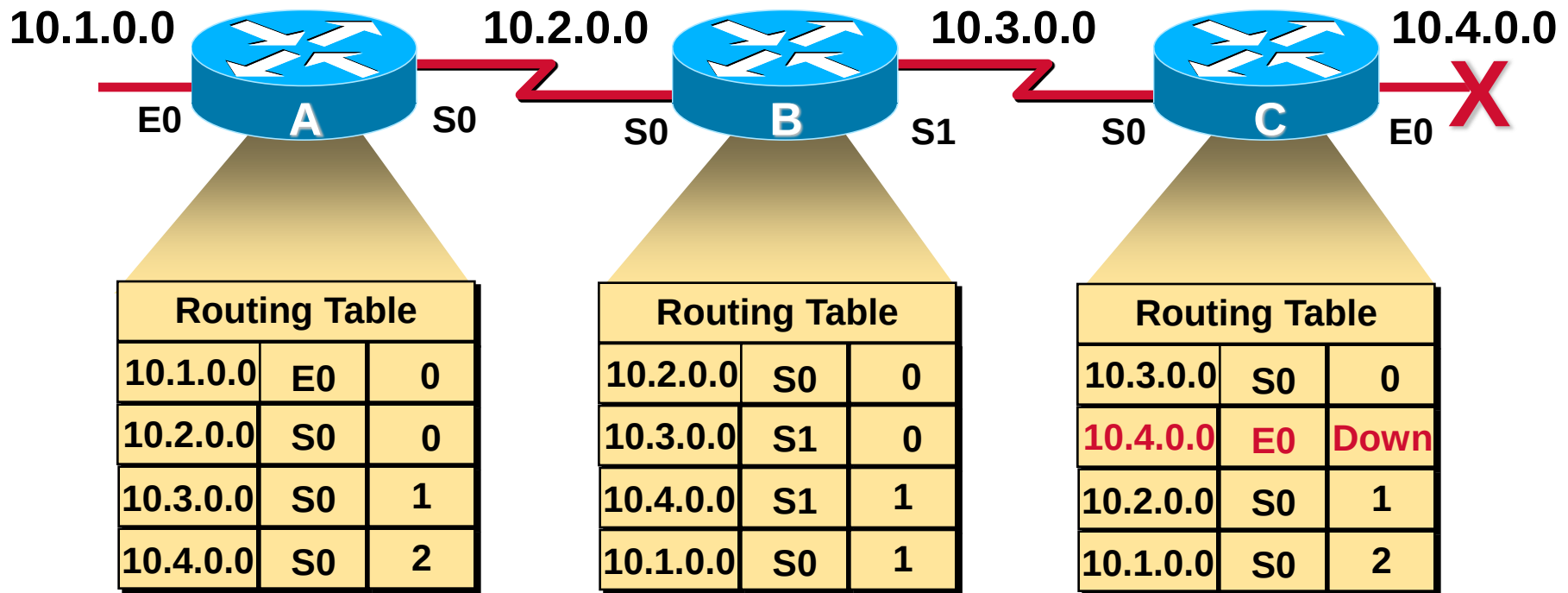
**Updates proceed step-by-step
from router to router**

Maintaining Routing Information Problem—Routing Loops



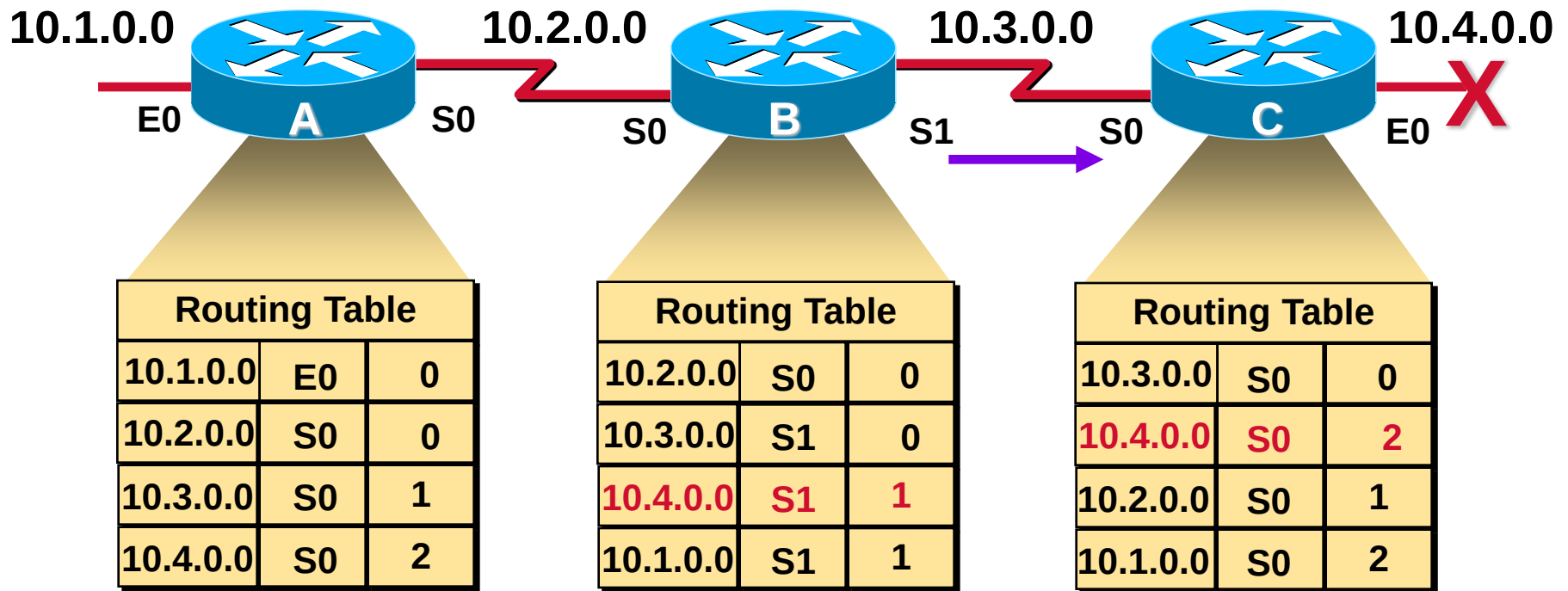
Each node maintains the distance from itself to each possible destination network

Maintaining Routing Information Problem—Routing Loops



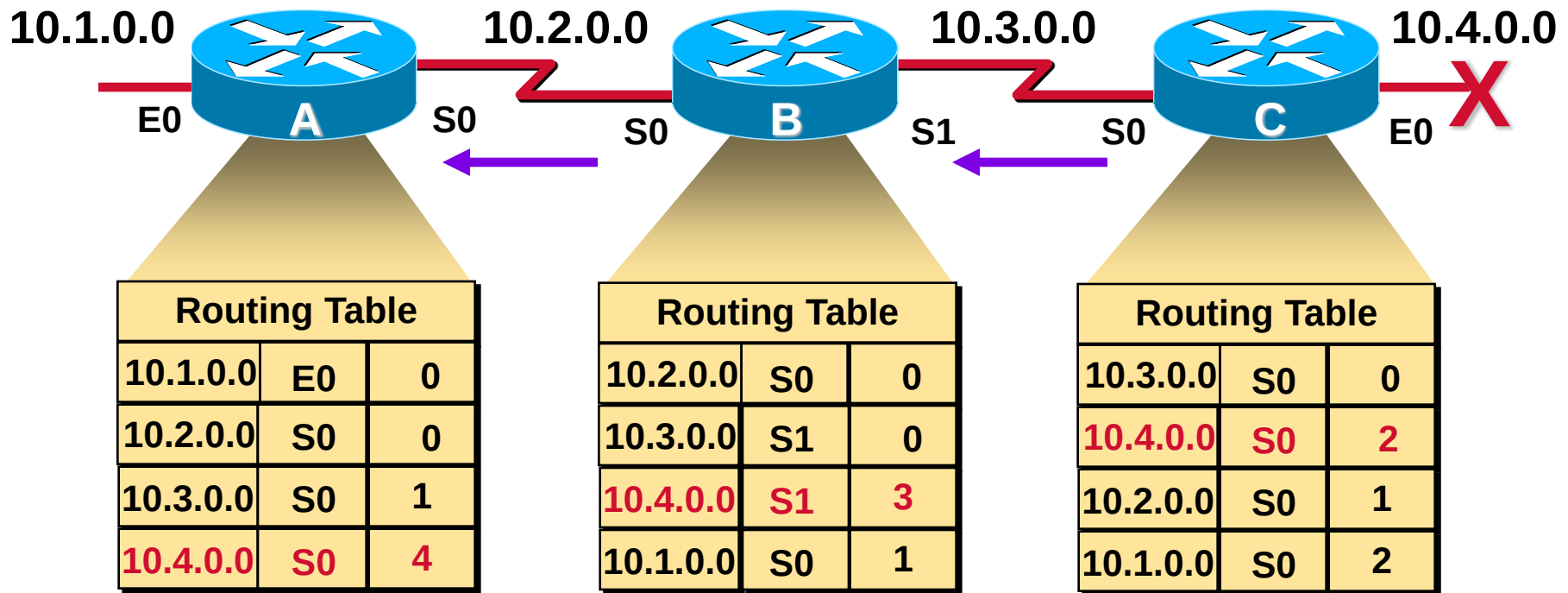
Slow convergence produces inconsistent routing

Maintaining Routing Information Problem—Routing Loops



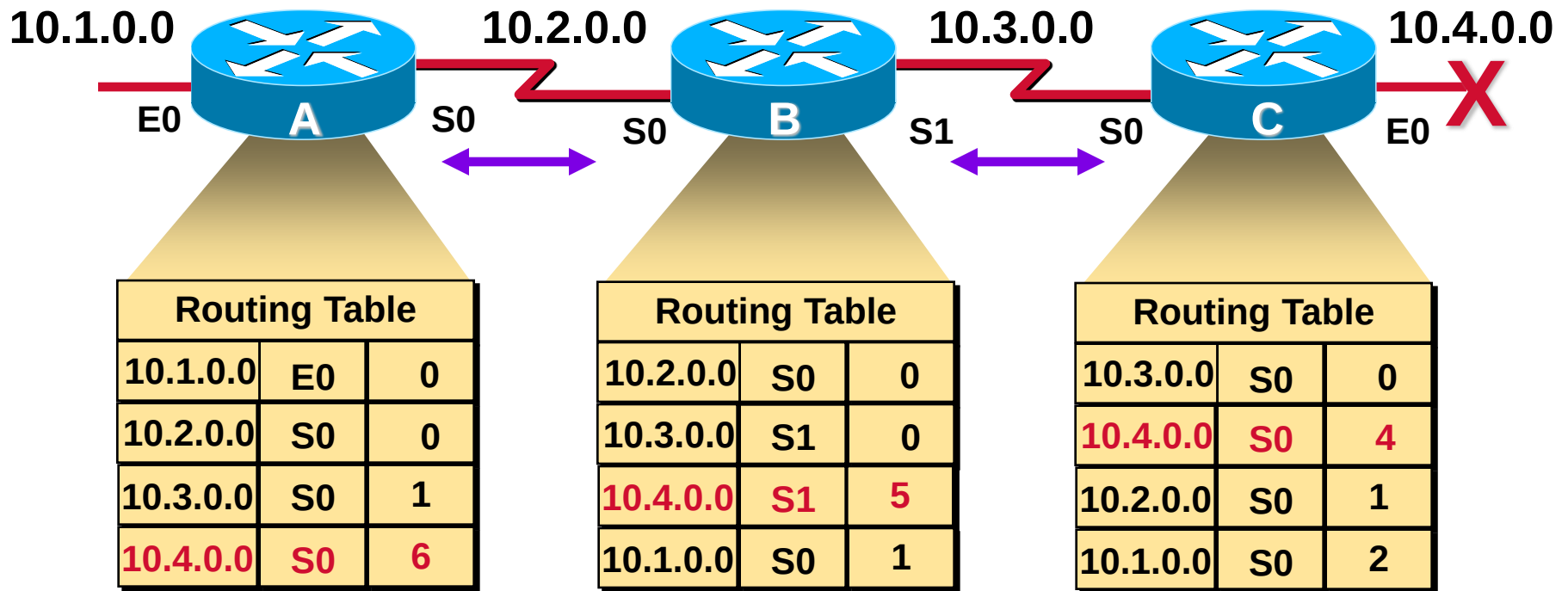
Router C concludes that the best path to network 10.4.0.0 is through Router B

Maintaining Routing Information Problem—Routing Loops



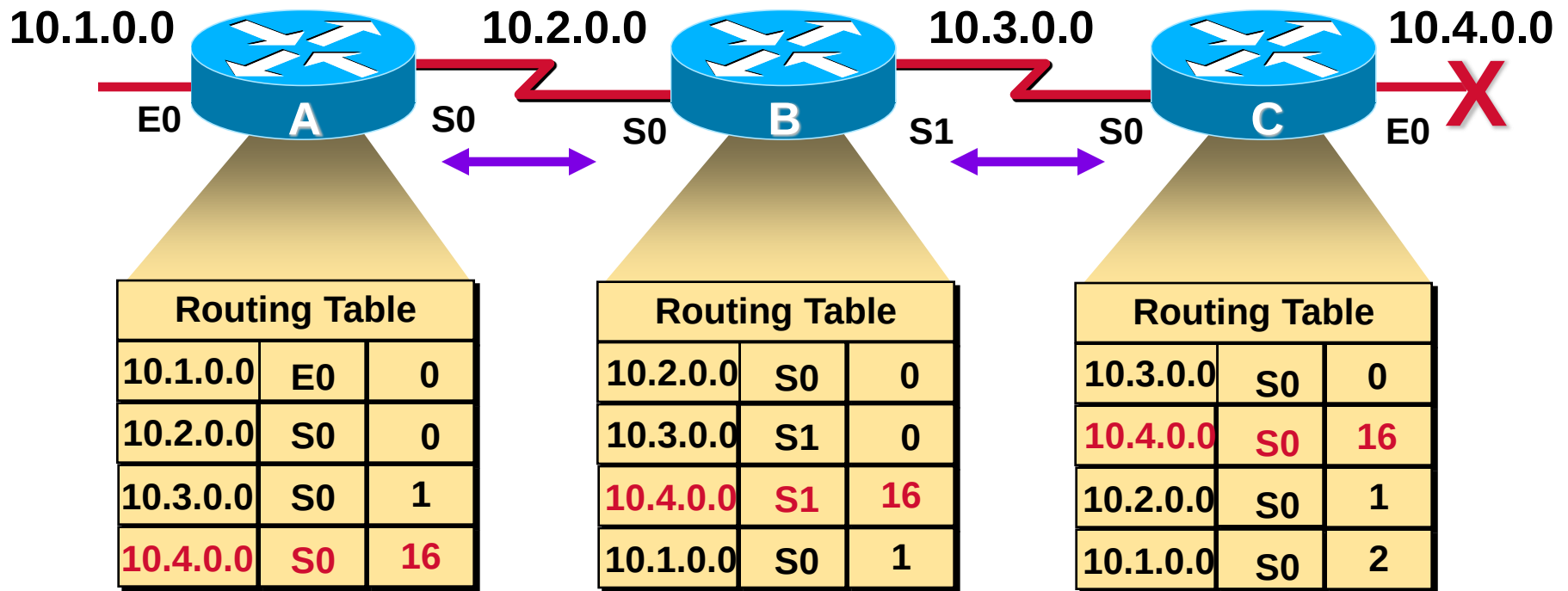
Router A updates its table to reflect the new but erroneous hop count

Symptom: Counting to Infinity



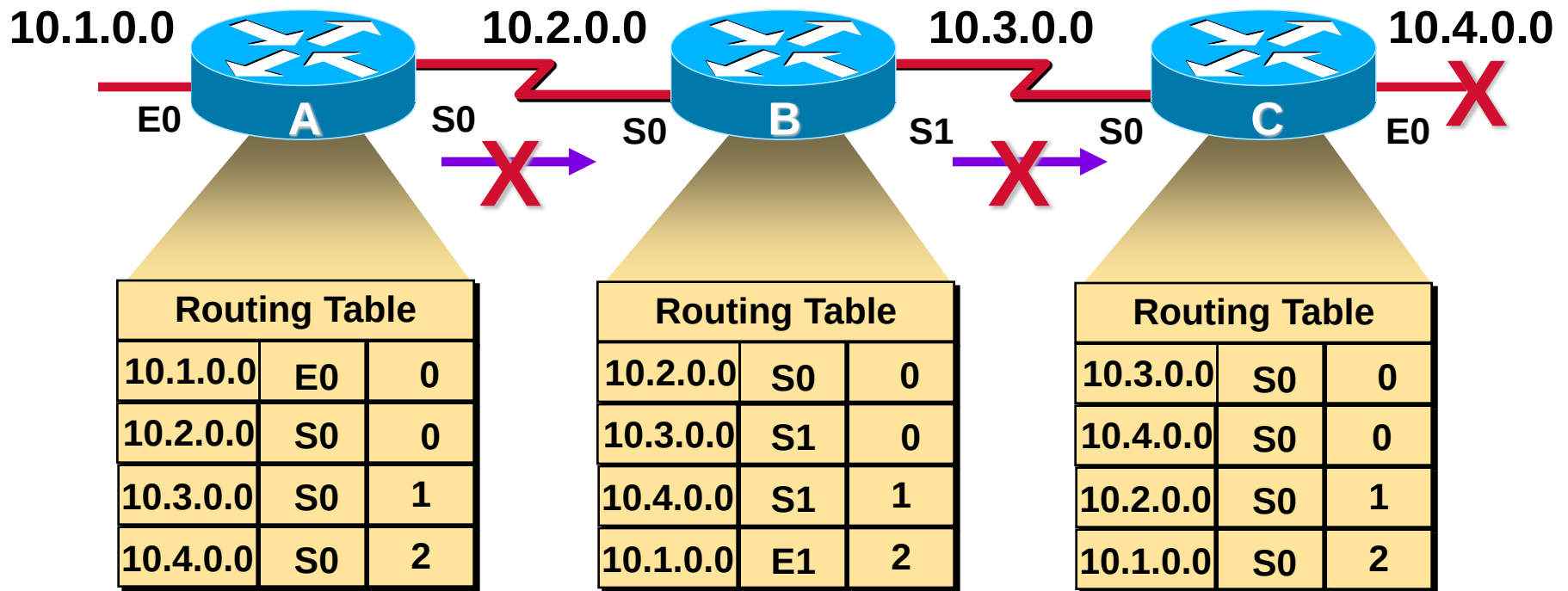
- Packets for network 10.4.0.0 bounce between routers A, B, and C
- Hop count for network 10.4.0.0 counts to infinity

Solution: Defining a Maximum



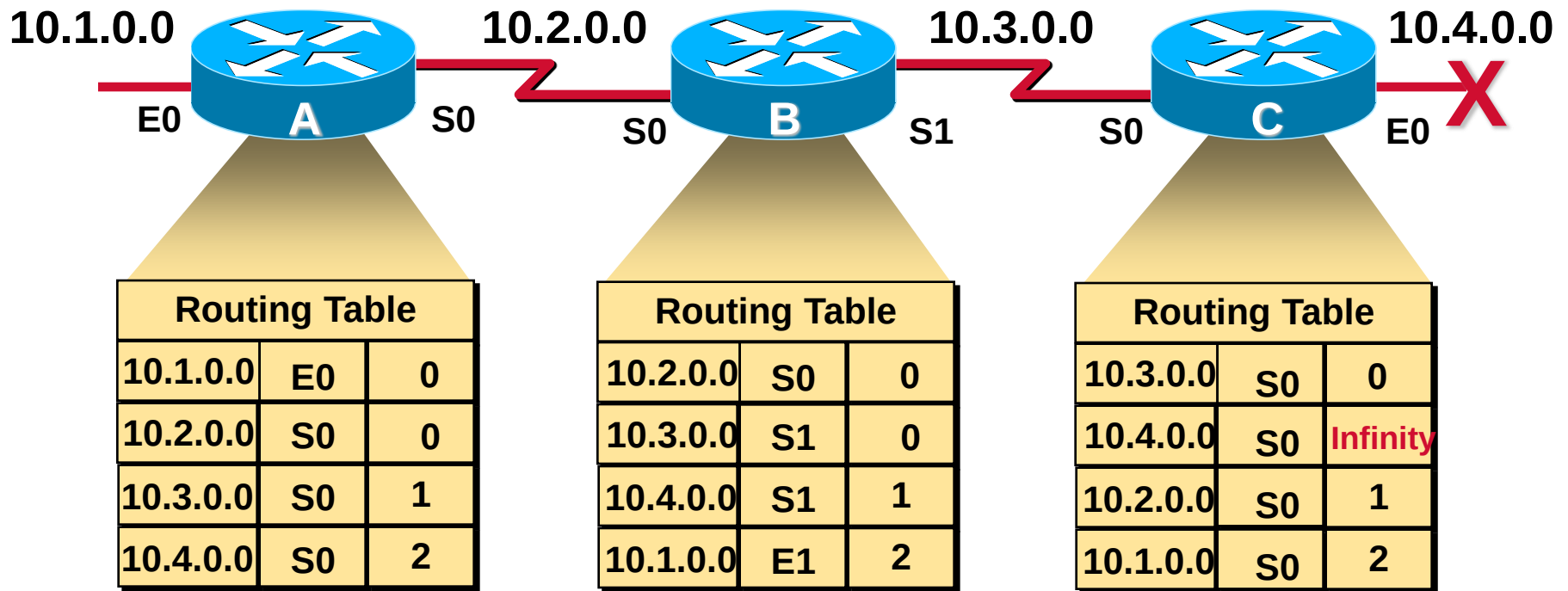
Define a limit on the number of hops to prevent infinite loops

Solution: Split Horizon



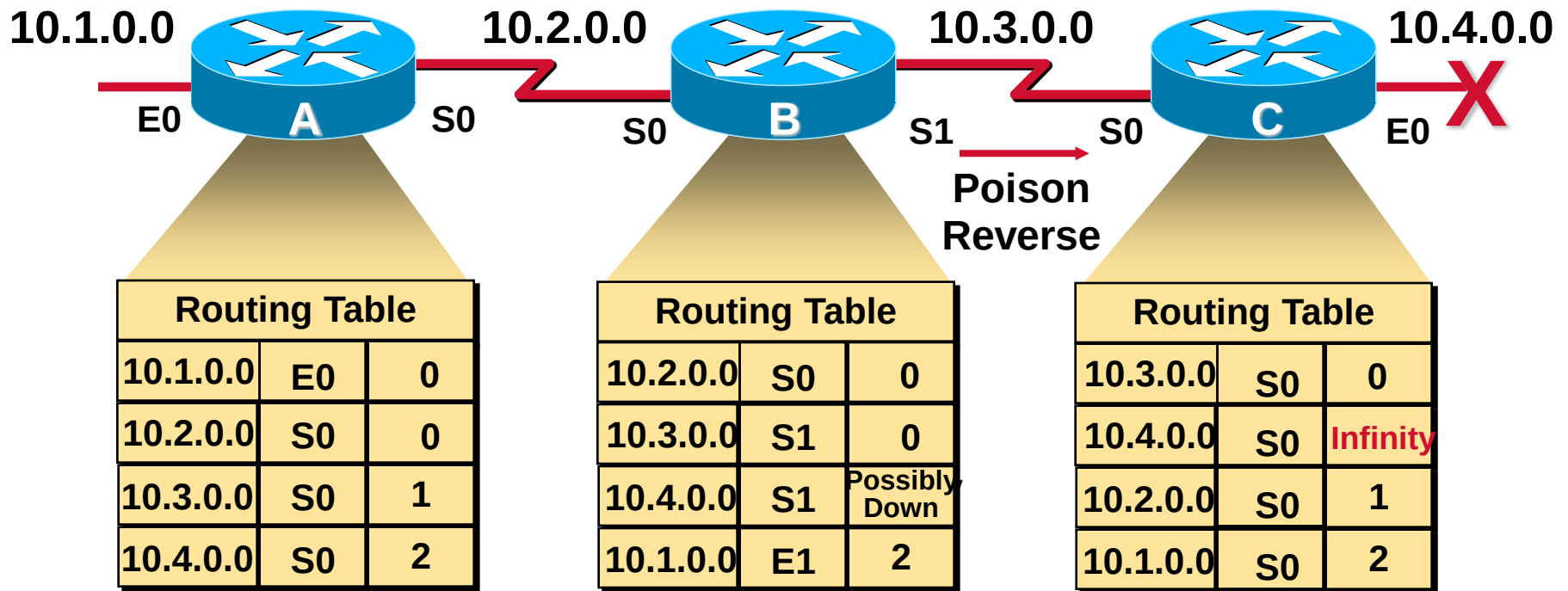
It is never useful to send information about a route back in the direction from which the original packet came

Solution: Route Poisoning



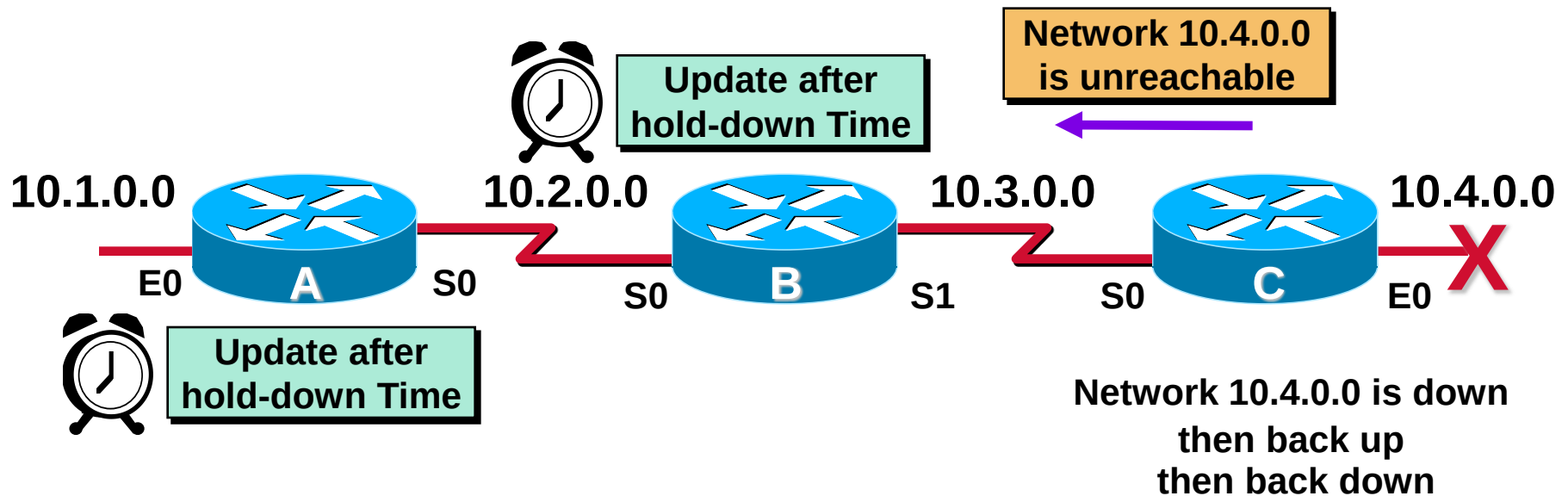
Routers set the distance of routes that have gone down to infinity

Solution: Poison Reverse



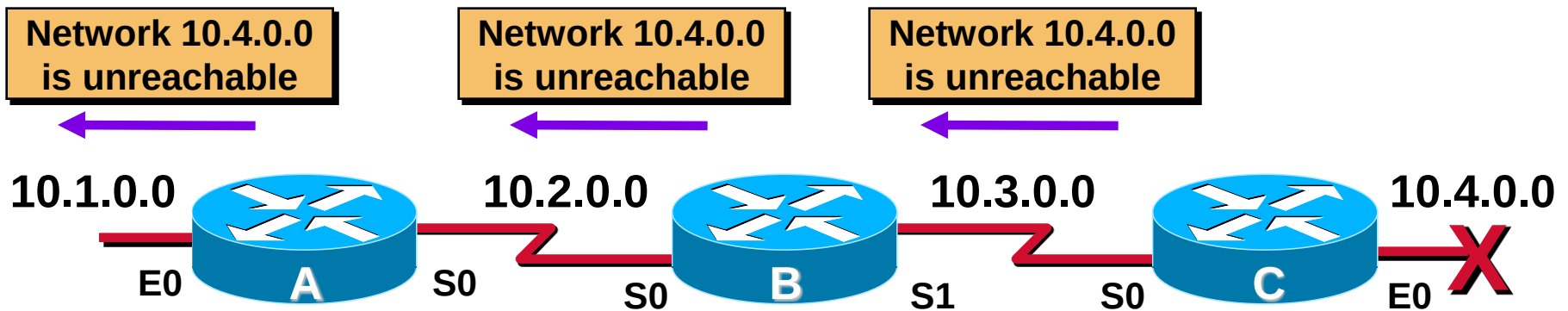
Poison Reverse overrides split horizon

Solution: Hold-Down Timers



Router keeps an entry for the network possibly down state, allowing time for other routers to recompute for this topology change

Solution: Triggered Updates



Router sends updates when a change in its routing table occurs

Dynamic Routing Configuration

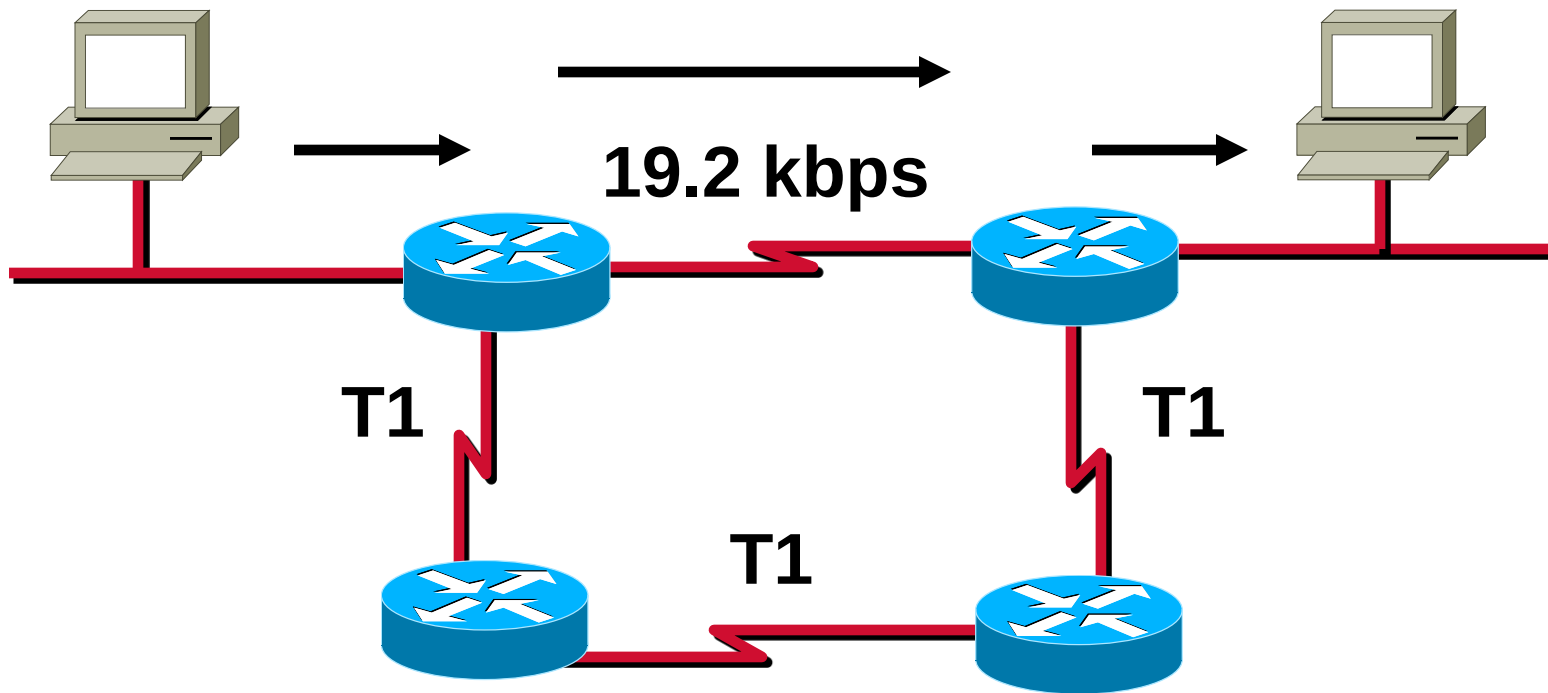
```
Router (config) #router protocol [keyword]
```

- Defines an IP routing protocol

```
Router (config-router) #network network-number
```

- Mandatory configuration command for each IP routing process
- Identifies the physically connected network that routing updates are forwarded to

RIP Overview



- Maximum six paths (default = 4)
- Hop count metric selects the path
- Routes update every 30 seconds

RIP Configuration

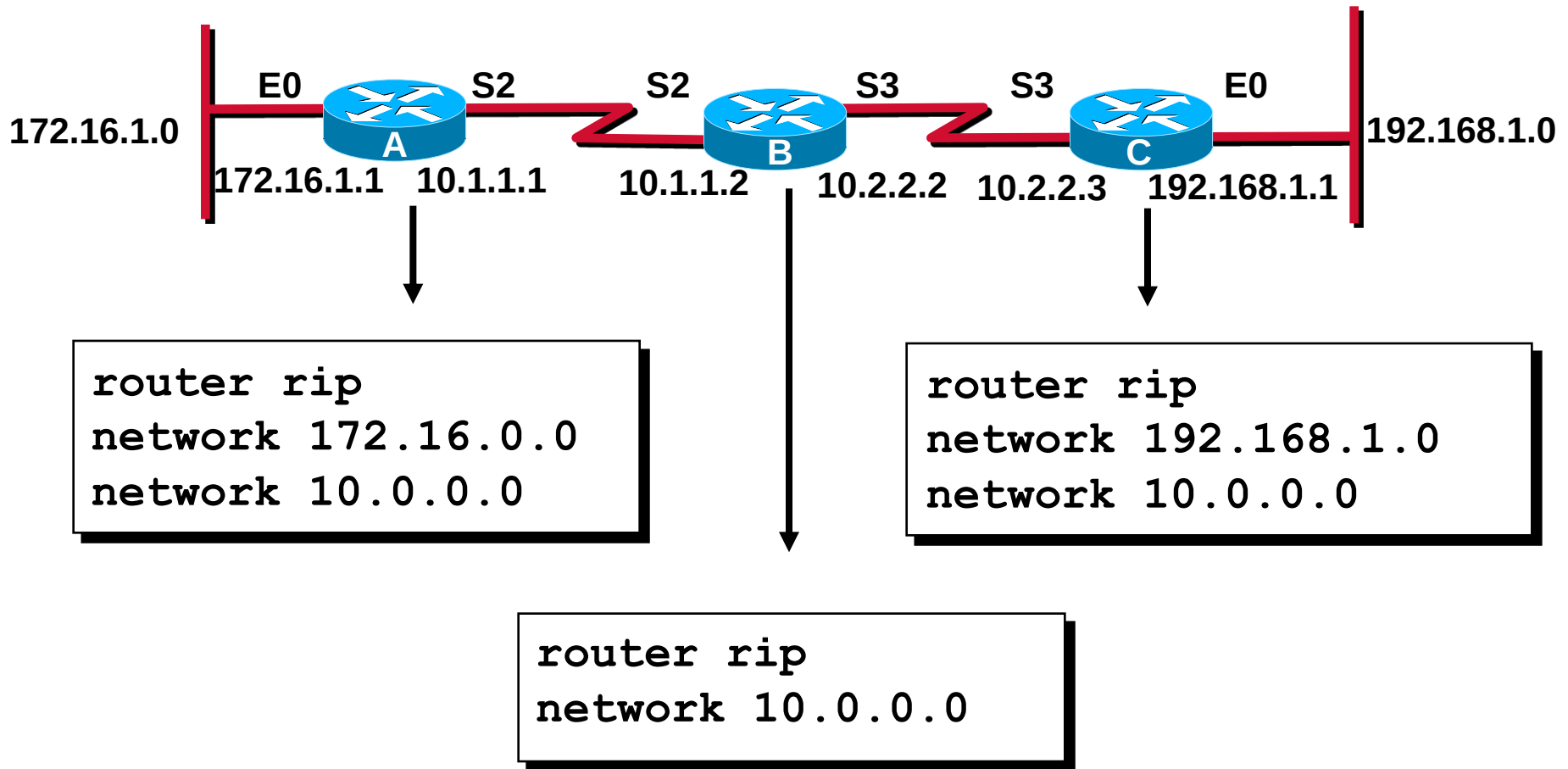
```
Router (config)#router rip
```

- Starts the RIP routing process

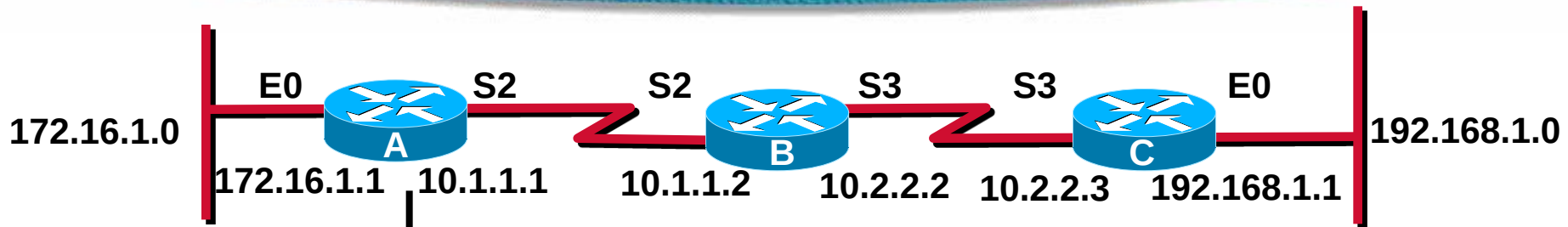
```
Router (config-router)#network network-address
```

- Selects participating attached networks
- The network number must be a major classful network number

RIP Configuration Example



Verifying the Routing Protocol—RIP



```
RouterA#sh ip protocols
Routing Protocol is "rip"
```

```
Sending updates every 30 seconds, next due in 0 seconds
```

```
Invalid after 180 seconds, hold down 180, flushed after 240
```

```
Outgoing update filter list for all interfaces is
```

```
Incoming update filter list for all interfaces is
```

```
Redistributing: rip
```

```
Default version control: send version 1, receive any version
```

Interface	Send	Recv	Key-chain
-----------	------	------	-----------

Ethernet0	1	1 2	
-----------	---	-----	--

Serial2	1	1 2	
---------	---	-----	--

```
Routing for Networks:
```

```
10.0.0.0
```

```
172.16.0.0
```

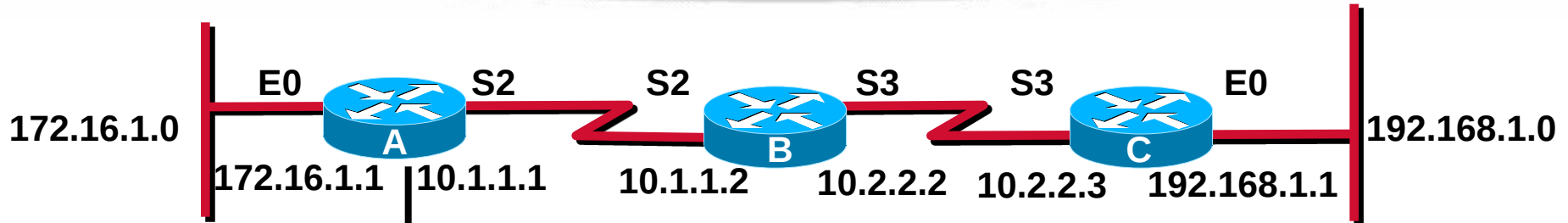
```
Routing Information Sources:
```

Gateway	Distance	Last Update
---------	----------	-------------

10.1.1.2	120	00:00:10
----------	-----	----------

```
Distance: (default is 120)
```

Displaying the IP Routing Table



```
RouterA#sh ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate
```

```
default
```

```
U - per-user static route, o - ODR
```

```
T - traffic engineered route
```

```
Gateway of last resort is not set
```

```
172.16.0.0/24 is subnetted, 1 subnets
```

```
C    172.16.1.0 is directly connected, Ethernet0
```

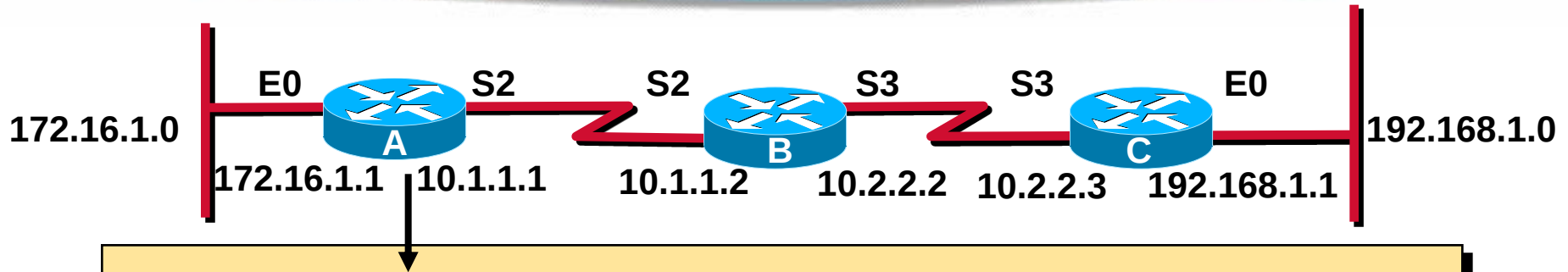
```
10.0.0.0/24 is subnetted, 2 subnets
```

```
R    10.2.2.0 [120/1] via 10.1.1.2, 00:00:07, Serial2
```

```
C    10.1.1.0 is directly connected, Serial2
```

```
R    192.168.1.0/24 [120/2] via 10.1.1.2, 00:00:07, Serial2
```

debug ip rip Command



```
RouterA#debug ip rip
RIP protocol debugging is on
RouterA#
00:06:24: RIP: received v1 update from 10.1.1.2 on Serial2
00:06:24:      10.2.2.0 in 1 hops
00:06:24:      192.168.1.0 in 2 hops
00:06:33: RIP: sending v1 update to 255.255.255.255 via
Ethernet0 (172.16.1.1)
00:06:34:      network 10.0.0.0, metric 1
00:06:34:      network 192.168.1.0, metric 3
00:06:34: RIP: sending v1 update to 255.255.255.255 via
Serial2 (10.1.1.1)
00:06:34:      network 172.16.0.0, metric 1
```